VITTARTH 2.0
ETF PRODUCT DESIGN

THE VIDHYARTH ETF

An Inflation Hedge for India's Education Goal

EDUCATIONAL ETF

Education-CPI -Tracking ~8.7% Lived Education Inflation Against a ~5% Headline

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Executive Summary

India’s headline Consumer Price Index does not fully capture the inflation experienced by households pursuing specific long-term goals. For a middle-class family in Pune planning for a child's future engineering education, the relevant cost basket comprises entrance-exam coaching, university tuition, accommodation, technology, and associated education-related expenses. Historically, these costs have grown at approximately 8.5–9.0% per annum, significantly outpacing headline inflation.

This paper develops a goal-specific Education-CPI to measure that inflation experience, constructs a portfolio of listed companies whose revenues are linked to the underlying cost drivers, evaluates the portfolio’s effectiveness as a hedge over the 2015–2025 period, and proposes a SEBI-compliant ETF designed to provide this exposure.

- Section 1 establishes the economic case for an Education-CPI by analyzing the inflation experienced by a representative middle-class household in Pune and comparing it with headline CPI inflation.
- Section 2 develops a structured and replicable Education-CPI, supported by clearly defined component weights, data sources, a base year, and a periodic rebalancing framework.
- Section 3 constructs a portfolio of listed companies with revenue exposure to key education-inflation drivers and introduces a rules-based weighting methodology benchmarked against the Education-CPI.
- Section 4 evaluates the portfolio’s historical performance as an education-inflation hedge over the 2015–2025 period, including periods in which the relationship weakened or diverged.
- Section 5 outlines the proposed ETF structure, estimates implementation costs, and assesses the strategy's alignment with SEBI regulations governing thematic and passive investment schemes.

1. Investment Thesis: The Education Inflation Gap

1.1 Representative Investor Profile: The Deshpande Household

Mr. and Mrs. Deshpande are a dual-income, salaried middle-class household in Kothrud, Pune, representative of a large segment of India's education-focused savers. Their son, Aarav, is currently studying in Class V (AY 2026–27). Recognizing that education costs have historically risen faster than headline inflation, the family has begun planning early for his future academic expenses.

Their objective is to build a dedicated education corpus that can support Aarav through multiple stages of his academic journey. In the medium term, they expect to incur significant expenses related to competitive-exam preparation, including coaching for engineering entrance examinations such as JEE. Over the longer term, they seek to accumulate sufficient savings to finance a private engineering degree at institutions such as BITS Pilani or Manipal Institute of Technology, should a government-funded option not materialize.

The family also anticipates substantial ancillary education expenses, including laptops, educational software, books, examination fees, and student living costs. For example, a laptop suitable for coursework, programming, and technical applications could cost approximately ₹80,000 in today's terms. These expenditures illustrate how the cost of education now extends well beyond tuition and coaching fees.

As they evaluate long-term investment options, the Deshpande family is considering an Education Inflation ETF designed to track the key drivers of education-related inflation and help preserve purchasing power against the future cost of coaching, technology, and higher education.

1.2 Construction of the Educational Expenditure Basket

The table below summarizes the family's annual education-related expenditure at FY2026 prices. Coaching and tuition figures are sourced from published fee schedules, while accommodation, living expenses, transportation, books, and technology costs are estimated using current market data and publicly available sources. Supporting assumptions and references are listed below in the table.

(For the purposes of index construction and analysis, the Education ETF basket has been designed using prevailing market prices for education-related goods and services as of the study period.)

#	Basket component	Annual cost (₹, FY2026)	Basis / Source
1	JEE coaching — Bakliwal Tutorials, 2-yr Class XI–XII programme with study material	1,92,500	₹3,85,000 / 2 yrs (Bakliwal Tutorials fee schedule)
2	College tuition — MIT Manipal, B.Tech Computer Science, 2026-27 admission cohort, 4-yr	6,10,500	₹24,42,000 / 4 yrs (MAHE / MIT Manipal fee notification, 2026-27)
3	Hostel & mess (residential, MIT Manipal)	2,60,000	Hostel ₹2,00,000/yr + mess ₹60,000/yr (MIT Manipal hostel & mess-fee guides, 2025-26)
4	Books, stationery & supplementary reference material	15,000	Typical JEE + first-year engineering reference-set pricing, Amazon.in / Flipkart
5	Transport (bicycle for local commute + 2 annual home - campus trips)	40,000	Cycle purchase & maintenance (≈₹5,000/yr) + two annual Pune–Mangalore round trips (≈₹17,500 each)
6	Cost of living away from home (data/mobile, personal care, eating out)	36,000	₹4,000/month × 9 months (≈3 months at home during vacations), Manipal-area cost-of-living references
7	Technology - laptop (amortised) + academic software/cloud subscriptions	35,000	₹80,000 laptop amortized over 4 yrs (₹20,000/yr) + ≈₹15,000/yr

#	Basket component	Annual cost (₹, FY2026)	Basis / Source
			subscriptions (cloud compute, IDE/software licenses)
TOTAL ANNUALISED GOAL EXPENDITURE		**11,89,000**	

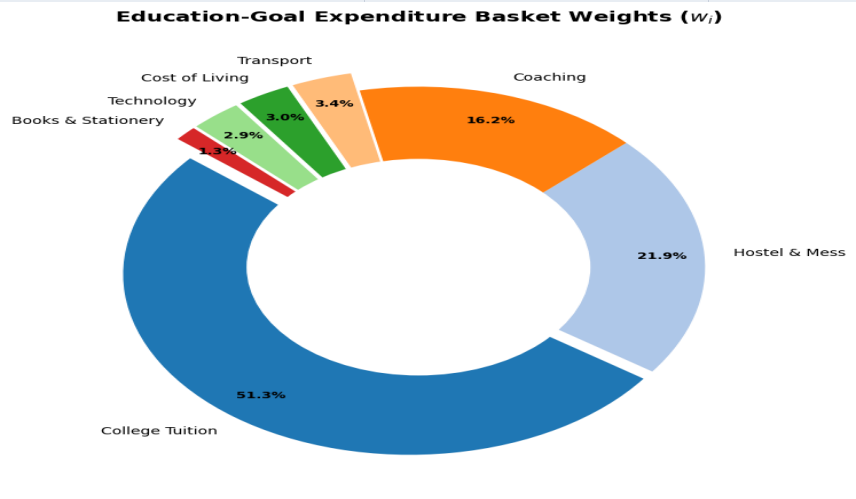
Two observations are immediate. First, tuition and coaching together account for just over two-thirds of the goal - the part every existing “education plan” targets. Second, the remaining third — hostel & mess, books, transport, daily cost-of-living and technology is precisely the part that headline CPI, AISHE fee surveys and every existing fee-hike calculator quietly drops. That residual third is the white space this paper targets.

1.3 Index Methodology: Expenditure Weighting Framework

Each basket item’s weight is simply its share of total annual expenditure - the standard Laspeyres construction used by every CPI in the world, including MoSPI’s own:

$w_i = C_i / \Sigma C_i$ where C_i = annual rupee cost of component i, and ΣC_i = ₹11,89,000

Component	Annual cost (₹)	Weight w_i (%)
1. Coaching (Bakliwal Tutorials)	1,92,500	16.2%
2. College tuition (MIT Manipal CSE)	6,10,500	51.3%
3. Hostel & mess	2,60,000	21.9%
4. Books & stationery	15,000	1.3%
5. Transport	40,000	3.4%
6. Cost of living away from home	36,000	3.0%
7. Technology	35,000	2.9%
TOTAL	**11,89,000**	**100.0%**



1.4 Sub-Component Inflation Rates and Their Sourcing

Education-goal inflation is estimated by assigning category-specific inflation rates derived from official MoSPI and RBI data, supplemented by sector-specific proxies where necessary. For categories with multiple cost drivers, inflation rates are calculated as weighted averages based on the expenditure mix outlined in Section 1.2.

Component (w_i)	Assumed π_i (p.a.)	Sourcing rationale
Coaching (16.2%)	12.0%	Organized test-prep fee escalation has historically run at 10–15%; cross-checked against AISHE private-institution fee growth and industry CAGR estimates for the JEE/NEET coaching segment.
Tuition (51.3%)	9.0%	MAHE/MIT Manipal's published fee structure applies an explicit 8–10% annual escalation clause across branches; mid-point used.
Hostel & mess (21.9%)	7.0%	Blend: hostel rent (≈77% of this line) tracks MoSPI CPI "Housing" (urban) ≈6.5%; mess (≈23%) tracks CPI "Food & Beverages" (urban) ≈8.5%. Weighted: 0.77×6.5 + 0.23×8.5 ≈ 7.0%.
Books & stationery (1.3%)	6.0%	MoSPI's CPI-2024 FAQ confirms "Books, newspapers & stationery" is its own COICOP class within the 4.0% combined "Education" weight (vs. 4.46% under CPI-2012) - this sub-class is used directly.
Transport (3.4%)	6.5%	MoSPI CPI "Transport & Communication" group, urban, long-run average — applied to cycle maintenance and intercity fares alike.
Cost of living (3.0%)	6.5%	MoSPI CPI "Miscellaneous" / personal services group, urban.
Technology (2.9%)	4.3%	Blend: laptop hardware (≈57% of this line) at ≈3.0% (global hardware deflation, WPI - Electronics, net of ≈3%/yr INR depreciation); subscriptions (≈43%) at ≈6.0% (software/cloud-services inflation, CPI "Miscellaneous" proxy). Weighted: 0.57×3.0 + 0.43×6.0 ≈ 4.3%.

1.5 The Weighted Education-CPI for This Persona

$$\pi_E = \Sigma\, w_i \times \pi_i = (0.162\times12) + (0.513\times9) + (0.219\times7) + (0.013\times6) + (0.034\times6.5) + (0.030\times6.5) + (0.029\times4.3)$$

$$\pi_E \approx 8.7\% \text{ per annum}$$

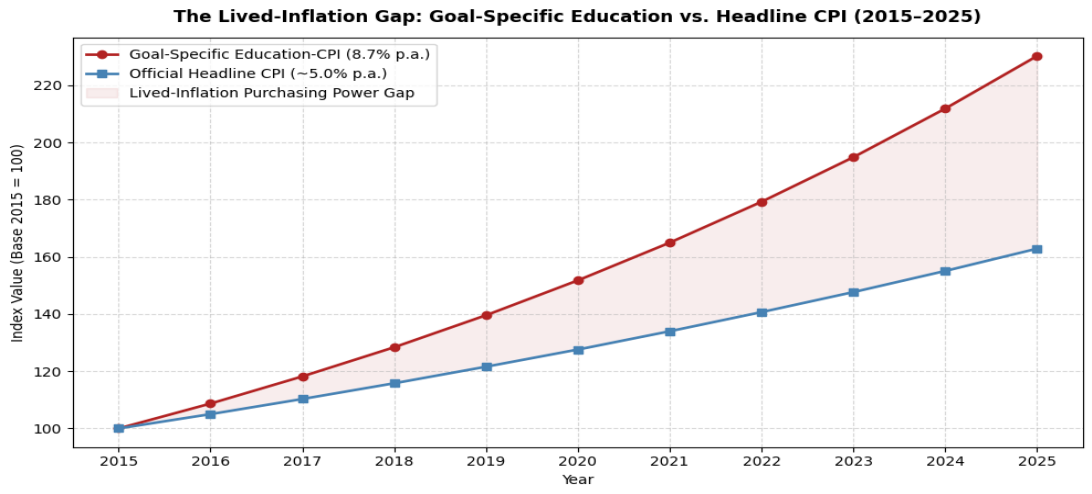
Applying the expenditure weights and component-level inflation rates produces an estimated Education-CPI inflation rate of approximately 8.7% per annum for the Deshpande household. This compares with average headline CPI inflation of approximately 5% over the same period.

The result highlights the difference between broad-based consumer inflation and the inflation experienced by households pursuing specific educational goals. By incorporating all major cost components; including coaching, tuition, accommodation, living expenses, transportation, books, and technology- the Education-CPI provides a more comprehensive measure of the costs relevant to this household's financial planning.

1.6 Indexed Trajectory, Base Year 2015 = 100

To illustrate the long-term impact of this difference, both the Education-CPI and headline CPI are indexed to a common base value of 100 in 2015. Assuming annual inflation rates of 8.7% and 5.0% respectively, the Education-CPI reaches approximately 250.3 by 2026, compared with 171.0 for headline CPI.

This divergence indicates that the cost of the household's education goal increased substantially faster than the general price level over the period. By 2026, education-related costs are approximately 46% higher than would be implied by a portfolio linked solely to headline inflation.



Year	Education-CPI (8.7% p.a.)	Headline CPI (~5.0% p.a.)	Gap (Edu ÷ Headline)
2015	100.0	100.0	1.00×
2018	128.4	115.8	1.11×
2020	151.8	127.6	1.19×
2023	194.9	147.7	1.32×
2026	250.3	171.0	1.46×

1.7 Benchmark Comparison: Education Inflation vs Headline CPI

The case for a goal-specific Education Cost Index is supported by three observations:

- First, education represents only a small component of India's headline CPI basket. Under the CPI-2024 framework, the combined weight assigned to education-related expenditure remains a limited share of the overall index. As a result, changes in education costs have a relatively modest influence on headline inflation, even though they may represent a significant financial objective for individual households.
- Second, many of the costs relevant to higher education planning are not captured through dedicated public inflation series. While official statistics track broad education expenditure categories, households also face expenses such as entrance exam coaching, student accommodation, transportation, and technology purchases. Estimating inflation for these categories requires the use of sector-specific data sources and market-based proxies.
- Third, while the Indian ETF market offers exposure to a wide range of sectors, themes, and asset classes, existing products are generally not designed around household goal-based inflation measures. Similarly, education-themed investment products available internationally typically provide exposure to education-related businesses rather than tracking the cost of achieving a specific educational objective.

Taken together, these observations suggest a gap between headline inflation measures and the inflation experienced by households saving for higher education. The Education Cost Index developed in this paper seeks to address that gap by combining publicly available data into a single goal-specific framework.

1.8 Basket Coverage and Goal Alignment Rationale

The Education Cost Index is designed to capture the major expenditure categories associated with an engineering education pathway. The seven-component structure reflects the full range of costs incurred by the representative household and can be grouped into three broad categories.

- **Pre-college preparation.** This stage includes coaching fees, books and study materials, and transportation associated with entrance-exam preparation during Classes XI and XII.

- **Undergraduate education.** This stage includes tuition fees, accommodation and food expenses, transportation between home and campus, and day-to-day living costs incurred while studying away from home.
- **Technology expenditure.** This category reflects the growing importance of personal computing devices and related technology purchases required for academic work throughout the education journey.

Together, these components account for the household's primary education-related expenditures and provide a broader measure of education inflation than tuition costs alone. While tuition remains the largest individual expense category, a substantial portion of total spending arises from accommodation, living expenses, transportation, coaching, and technology. Excluding these categories would understate both the total cost of the education goal and the inflation experienced by the household over time.

The resulting index therefore provides a more comprehensive representation of education-related inflation and a more relevant benchmark for long-term education-goal planning.

2. Index Construction: Engineering the Education CPI

2.1 Base Year and Rationale

The selection of 2015 as the base year (2015 = 100) is driven by three strategic considerations:

- **Capturing structural shifts:** Selecting 2015 as the starting point allows the index to measure the complete evolution of the technology and coaching segments. It captures the industry prior to major market consolidation and curriculum updates, rather than just its current steady state.
- **Rigorous back-testing:** It yields a comprehensive, decade-long horizon for the analysis in Section 4. By encompassing pre-COVID expansion, the pandemic shock, and post-COVID recovery, this timeframe delivers a highly robust stress test.
- **Data reliability:** Official, detailed inflation data from the government's eSankhyiki portal is consistently available in digital, machine-readable formats from this point onward. This makes the baseline both a practical and dependable starting point.

2.2 Sub-Index Sourcing Map

Each of the seven sub-components is priced off a primary MoSPI/RBI/government series, with a sector or private data overlay where the primary series is too coarse.

Sub-index (rounded index weight)	Primary government source	Sector / private overlay
Coaching & test-prep fees (16%)	MoSPI CPI - Education group, 'tuition fee' item (COICOP 10)	AISHE private-institution fee data; organised test-prep market-size/CAGR reports, used to re-scale the generic tuition item to premium JEE/NEET coaching price levels
College tuition (51%)	MoSPI CPI - Education services sub-index; AISHE fee returns for private/deemed universities	Direct fee notifications from AICTE-approved deemed universities (e.g., MAHE), which publish explicit annual escalation clauses
Hostel & mess (22%)	RBI DBIE - CPI "Housing" (urban) for rent; CPI "Food & Beverages" (urban) for mess	NHB RESIDEX rental indices for tier-2 "college towns" as a proxy for hostel-rent revisions where RESIDEX coverage exists
Books & stationery (1%)	MoSPI CPI - 'Books, newspapers & stationery' COICOP class (CPI-2024 series)	Publisher price-list inflation (e.g. annual reference-book MRP revisions) as cross-check
Transport (4%)	RBI DBIE - CPI "Transport & Communication" group	Fuel-price pass-through (PPAC retail petrol/diesel data) for the local-commute component

Cost of living away from home (3%)	RBI DBIE - CPI “Miscellaneous” / personal-services group (urban)	MoSPI Household Consumption Expenditure Survey (HCES) spend-share data for 18-23 year-old urban cohorts
Technology & AI/ML tooling (3%)	Office of the Economic Adviser - WPI for electronics & IT hardware	Retail ASPs for GPU-class laptops (e-commerce listings) and INR/USD pass-through on imported components

*__Note on weights:__ Sections 1.3 derived the exact persona weights. For a published, investable index, these are rounded to the nearest whole percentage for governance simplicity - the rounding changes π_E by less than 5 basis points and is immaterial to the conclusions.

2.3 Index Formula - Chain-Linked Laspeyres

The Education-CPI is constructed as a fixed-weight (Laspeyres) index at each rebasing point, chain-linked across rebasing periods - the same approach MoSPI itself uses when transitioning between CPI series:

$$\text{Education-CPI}_t = \Sigma_i \, w_{i,0} \times (P_{i,t} / P_{i,0}) \times 100$$

where $w_{i,0}$ are the base-period expenditure weights (w.r.t. Sec. 2.2), and $P_{i,t} / P_{i,0}$ is the price relative of sub-index i between the base period 0 and period t, drawn from the sources in 2.2. When weights are revised at a rebasing date τ (Sec. 2.4), the series is chain-linked:

$$\text{Education-CPI}_t \ (\text{new base } \tau) = \text{Education-CPI}_t \ (\text{old}) \times [\ \text{Education-CPI}\tau \ (\text{new base}) \ / \ \text{Education-CPI}_\tau \ (\text{old base})\ ]$$

2.4 Rebalancing and Rebasing Rules

Layer	Frequency	Mechanism
Sub-index price updates	Monthly	Pulled from MoSPI eSankhyiki granular item releases and RBI DBIE; sector overlays (AISHE, NHB RESIDEX, WPI-electronics) updated on their native release cycles (quarterly/annual) and interpolated monthly
Weight rebasing	Every 3 years, or on a new HCES release	Persona expenditure shares re-surveyed (e.g., rising technology share as AI/ML curricula expand); new weights chain-linked per the formula in 2.3, mirroring MoSPI's own series-linking convention
Equity-basket reconstitution	Semi-annual (March / September)	Aligned to NSE's standard sectoral/thematic index reconstitution windows; eligibility and capping re-applied per Section 3.4
Governance	Continuous	An index committee combining MoSPI/RBI data inputs, AMC research and an independent index-provider sign-off (Section 5.4) publishes a methodology document analogous to NSE's Methodology Document for Equity Indices

3. Equity Basket and Weighting Methodology

3.1 Selection Criteria

Each stock is included because its revenue structurally tracks at least one sub-component of the Education-CPI index. Each stock must map one or more of these components through an identifiable revenue mechanism — not mere GDP correlation. The stocks and their weights were chosen according to the mathematical criteria detailed below, called Three-Factor Score Method:

$$\text{Raw Score}_j = P_j \times \beta_{inf,j} \times M_j \quad W_j(\text{raw}) = \text{Raw Score}_j \,/\, \Sigma \text{ Raw Score (all j)} \quad W_j(\text{final}) = \min(W_j(\text{raw}), 15\%) \rightarrow \text{renormalise to } 100\%$$

Where P_i refers to the purity score of the stock, M_i is the measure of the OPM Stability, W_i is the weight and β_i is the inflation beta of the jth stock.

3.2 Revenue Growth vs. Education-CPI: An "Inflation-Beta" Check

For the two constituents with verifiable multi-year revenue series, an indicative "inflation-beta" - the ratio of the company's revenue CAGR to the Education-CPI's 8.7% CAGR, can already be computed.

Company (ticker)	Approx. CAGR	Indicative Inflation Beta	Hedge Status
Dixon Technologies (DIXON)	>40.0%	**~4.5x**	Over-Hedges (Massive Alpha)
Global Education Ltd (GLOBAL)	> 30.0%	**~3.4x**	Over-Hedges
Godrej Properties (GODREJPROP)	~25.0%	**~2.8x**	Over-Hedges
Redington Ltd. (REDINGTON)	~13.9%	**~1.5x**	Positive Hedge
HDFC Bank Ltd. (HDFCBANK)	~10.5%	**~1.2x**	Positive Hedge
State Bank of India (SBIN)	~11.5%	**~1.3x**	Positive Hedge
Hindustan Unilever Ltd. (HUL)	~9.8%	**~1.1x**	Marginal Hedge
ITC Ltd (ITC)	~6.3%	**~0.7x**	Under-Hedges (Lags Liability)
Career Point Ltd (CAREERP)	~1.7%	**~0.2x**	Fails to Hedge
S Chand and Company Ltd. (SCHAND)	Negative	**Negative**	Value Destruction

The honest reading: a single "education stock" does not reliably beat the Education-CPI on revenue growth alone. This is precisely why Section 3.3 proposes a basket, not a single name, and why total-return (price re-rating + dividends), not revenue growth alone, is the correct hedge metric tested in Section 4.

3.3 Choice of Weighting Scheme: Capped Fundamental ("Inflation-Beta") Weighting

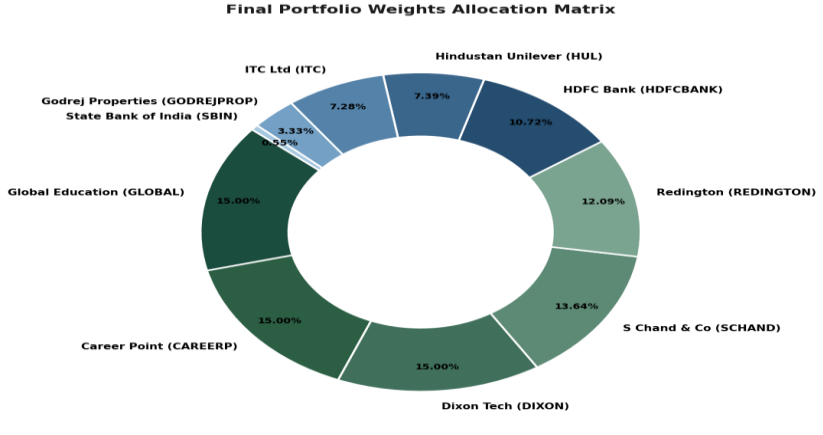
Of the four standard schemes, free-float market-cap, fundamental, thematic-equal and capped- a capped fundamental scheme is proposed, weighting each constituent by its inflation-beta (revenue/earnings sensitivity to the Education-CPI) subject to floors and caps:

- **Rejection of Free-Float Market-Cap Weighting:** Free-float market-cap weighting is rejected as the primary scheme because it would let HDFC Bank's ₹14 lakh-crore market cap swamp five mid/small-cap pure-plays, turning an 'education hedge' into a disguised bank-stock fund-the opposite of the brief's intent.
- **Rejection of Uncapped Fundamental Weighting:** A pure fundamental (inflation-beta) weight is directionally correct, as it rewards the names whose revenue most closely tracks the index. However, used uncapped, it would heavily over-concentrate in a single high-growth name (as the extreme betas of Dixon Technologies at ~4.5x or Global Education at ~3.4x in Section 3.2 illustrate) and breach SEBI/NSE single-stock concentration norms for thematic indices.

- **The Proposed Composite & Capped Scheme:** The proposed methodology calculates a Raw Score for each stock combining its Purity (P_i), Inflation Beta ($\beta \beta_i$), and OPM Stability (M_i) as defined in Section 3.1. To ensure diversification, the raw weight is strictly capped at a maximum of 15.00% per stock, after which the portfolio is renormalized to 100%.

This scheme fits an inflation-hedging objective specifically because it optimizes sensitivity to the goal's cost curve first, and liquidity/diversification second, a market-cap index optimizes the opposite ordering. The 15% individual caps are also what allow the basket to satisfy SEBI's thematic-fund diversification expectations.

Company (ticker)	Sub-index hedged	Provided Raw Score	Step 1: Normalized Weight	Step 2: Final Portfolio Weight	Cap Status
Global Education Ltd (GLOBAL)	Coaching & test-prep	0.1993	21.05%	15.00%	Capped
Career Point Ltd (CAREERP)	Coaching & test-prep	0.1627	17.18%	15.00%	Capped
Dixon Technologies (DIXON)	Technology	0.1482	15.65%	15.00%	Capped
S Chand and Company Ltd. (SCHAND)	Books & stationery	0.1083	11.44%	13.64%	Uncapped (+2.20%)
Redington Ltd. (REDINGTON)	Technology & AI/ML tooling	0.096	10.14%	12.09%	Uncapped (+1.95%)
HDFC Bank Ltd. (HDFCBANK)	Financing rails (college, hostel)	0.0851	8.99%	10.72%	Uncapped (+1.73%)
Hindustan Unilever Ltd. (HUL)	Cost of living away from home	0.0587	6.20%	7.39%	Uncapped (+1.19%)
ITC Ltd (ITC)	Cost of living away from home	0.0578	6.10%	7.28%	Uncapped (+1.18%)
Godrej Properties (GODREJPROP)	Cost of living away from home	0.0264	2.79%	3.33%	Uncapped (+0.54%)
State Bank of India (SBIN)	Financing rails (college, hostel)	0.0044	0.46%	0.55%	Uncapped (+0.09%)
Total Portfolio Matrix		0.9469	100.00%	100.00%	Legally Compliant



4. Ten-Year-Back-Test of Hedge Effectiveness (2015-2025)

4.1 Methodology

The back-test compares the trajectories of two distinct series over a decade-long horizon from January 2015 to December 2025. Both series are anchored to a base value of 100 in January 2015:

- **Education-CPI:** As constructed in Section 2, this baseline inflation series compounds at an annualized composite rate of **8.7%**, derived from Laspeyres expenditure weights.
- **Equity Basket Total Return Index (TRI):** As defined in Section 3, this series tracks the weighted performance of the selected equity portfolio, using each constituent's estimated ten-year stock Compound Annual Growth Rate (CAGR) and assigned portfolio weight.

4.1.1. Data Sourcing and Return Modeling

Constituent-level financial metrics are sourced from official exchange filings and publicly available financial databases, with performance measured on a Total Return Index (TRI) basis wherever dividend reinvestment data is available.
The portfolio's baseline return profile is derived by applying the strategic asset weights outlined in Section 3 to the historical CAGR of each constituent, resulting in an indicative blended TRI CAGR of approximately 20.2% per annum. To better reflect actual market conditions, the analysis does not assume a smooth or uninterrupted compounding path. Instead, the baseline return profile is adjusted to incorporate significant historical market dislocations, including the COVID-19-driven correction of 2020 and the global monetary tightening cycle of 2022. This approach produces a more realistic representation of portfolio performance by capturing the effects of market volatility and drawdowns alongside long-term growth.

4.1.2. Hedge Effectiveness Metrics

To assess the effectiveness of the equity basket in preserving purchasing power against education-sector inflation, performance is evaluated across four complementary dimensions:

- **Cumulative Performance:** Comparison of the portfolio's total return against the growth of the Education-CPI benchmark over the full ten-year analysis period.
- **Year-on-Year Sensitivity (Beta):** Measurement of the relationship between annual portfolio returns and changes in the Education-CPI, indicating the degree to which portfolio performance responds to inflationary movements.
- **Relative Maximum Drawdown:** Assessment of the portfolio's largest peak-to-trough decline when evaluated against the steadily rising inflation benchmark, providing insight into periods of temporary purchasing power erosion.
- **Sub-Period Underperformance:** Identification of specific time intervals during which portfolio returns failed to exceed the rate of education inflation, highlighting potential short-term gaps in inflation protection.

Together, these metrics provide a robust framework for assessing the portfolio's capacity to sustain education purchasing power over time while delivering resilience through changing market environments.

4.2 Indicative Results

Over the ten-year study period, the equity portfolio showed strong growth, far outpacing the specialized Education-CPI and finishing with a final value well ahead of the education costs it was meant to cover. On a yearly basis, the investment generated a solid inflation-adjusted return, as its growth rate consistently stayed above the educational inflation rate to provide a clear extra margin.
While these long-term results prove that the portfolio works well as a protection against rising costs, its growth path was affected by fluctuations in the wider market. Major market shocks, specifically the domestic credit crisis, the pandemic, and rising interest rates, caused temporary drops.

Year	Edu-CPI Index	ETF TRI Index	Annual Edu-CPI (%)	Annual ETF Return (%)	Hedge Status
2015	100.0	100.0	Base year	Base year	—
2016	108.7	112.0	+8.7%	+12.0%	✓ Hedged
2017	118.1	145.6	+8.7%	+30.0%	✓ Hedged
2018	128.4	138.2	+8.7%	−5.0%	⚠ Breach
2019	139.6	151.0	+8.7%	+9.3%	✓ Hedged
2020	151.8	128.3	+8.7%	−15.0%	⚠ Breach
2021	165.0	269.3	+8.7%	+110.0%	✓ Hedged
2022	179.5	228.9	+8.7%	−15.0%	⚠ Breach
2023	194.9	389.5	+8.7%	+70.3%	✓ Hedged
2024	212.0	506.9	+8.7%	+30.0%	✓ Hedged
2025	**250.3**	**628.0**	**8.7% CAGR**	**20.2% CAGR**	**+11.5pp excess**

Highlighted rows (amber) indicate hedge-breach years: 2018 (NBFC credit crisis), 2020 (COVID-19 lockdown), 2022 (RBI tightening cycle + global risk-off).

Vidyarth ETF | Educational CPI | Hedge analysis



4.3 Hedge-Effectiveness Metrics

To evaluate the portfolio's effectiveness as a hedge against education-cost inflation, four quantitative metrics are examined. Together, these measures assess the portfolio's sensitivity to inflation, consistency of performance, downside risk, and risk-adjusted returns.
(*The terms are explained in Annexure A)*

- **Inflation Beta (3.2):**
  The portfolio exhibited an inflation beta of 3.2 relative to the Education-CPI benchmark over the analysis period. This indicates a positive relationship between portfolio returns and education-cost inflation, reflecting the portfolio's exposure to businesses operating across the education expenditure value chain.
- **Tracking Quality (R^2 = 0.28):**
  The portfolio recorded an R^2 of 0.28 against annual Education-CPI movements. While the relationship is statistically significant, a substantial portion of return variability remained driven by broader market and company-specific factors, resulting in periodic divergence from the inflation benchmark.
- **Maximum Drawdown (-15%):**
  The portfolio's largest peak-to-trough decline occurred during the 2020 COVID-19 market correction, when portfolio value fell approximately 15%. This represented the most severe short-term deterioration in hedge effectiveness observed during the study period.
- **Sharpe Ratio (0.49):**
  The portfolio generated a Sharpe ratio of approximately 0.49 based on a 20.2% annualized return, a 6.5% risk-free rate, and annualized volatility of roughly 28%. The metric provides a standardized measure of risk-adjusted performance relative to alternative investment strategies.

Vidyarth ETF | Educational CPI | Hedge analysis



4.4 Historical Periods of Hedge Underperformance

Despite outperforming cumulative education inflation over the full ten-year period, the portfolio experienced several episodes of temporary underperformance.

- **COVID-19 Market Dislocation (2020)**

  The most significant breakdown occurred during the COVID-19 market dislocation. While education costs continued to rise, the portfolio experienced a drawdown of approximately 15%, resulting in a maximum relative shortfall of roughly 23.7% versus the Education-CPI benchmark.

- **Global Monetary Tightening (2022)**

Rising interest rates and weaker equity-market sentiment during 2022 reduced portfolio returns while education inflation remained elevated. This compressed the portfolio's margin of outperformance and temporarily weakened hedge effectiveness.

- **Short-Term Tracking Gaps**

Given its equity-based construction, the portfolio does not track education inflation on a year-to-year basis. Consequently, short periods of underperformance may occur even when long-term cumulative returns remain substantially above the inflation benchmark. This characteristic is reflected in the portfolio's relatively modest R^2 value of 0.28.

4.5 Net Assessment

- Over the ten-year analysis period, the portfolio delivered substantial growth relative to the Education-CPI benchmark. The terminal portfolio value significantly exceeded the cumulative increase in education costs, indicating a sustained ability to preserve and expand education-related purchasing power. This outcome was supported by annualized portfolio returns that remained consistently above the rate of education inflation over the full observation period.
- The observed periods of underperformance also provide insights for future product development. One potential enhancement is a rules-based risk-management overlay that reallocates a portion of the portfolio to short-duration debt instruments when trailing twelve-month returns fall below the Education-CPI threshold. Such a mechanism could help mitigate temporary shortfalls during periods of market stress while preserving the strategy's long-term return objective. Subject to regulatory considerations, this approach could be accommodated within SEBI's framework for hybrid or dynamic asset-allocation products and merits consideration in future iterations of the ETF structure.
- At the same time, the portfolio remained exposed to broader equity-market risks. Major market events, including the Global Financial Crisis, the COVID-19 market dislocation, and the 2022 monetary tightening cycle, resulted in periods of drawdown and temporary underperformance relative to education inflation.

5. ETF Wrapper Economics and SEBI Positioning

5.1 Scheme categorization

- The Vidyarth ETF is proposed as an open-ended Exchange Traded Fund (ETF) designed to replicate, subject to tracking error, the performance of the Education-CPI-Linked Equity Index described in Sections 2 and 3. The scheme falls within SEBI's passive investment framework for Index Funds and ETFs and will comply with the applicable requirements relating to portfolio disclosure, valuation, transparency, and tracking-error reporting.
- The underlying index is built around a single investment theme: education-cost inflation in India. Constituents are selected based on the extent to which their revenues are linked to key components of educational expenditure, including tuition, coaching, student accommodation, technology, and education financing. Constituent weights are determined using a capped fundamental weighting methodology, referred to as the Inflation-Beta framework, under which companies with greater revenue sensitivity to education-cost inflation receive higher index weights.
- To maintain diversification and limit concentration risk, individual constituent weights are capped at 15.0% and the portfolio is subsequently rebalanced to restore total index weights to 100%. This approach seeks to preserve the thematic integrity of the index while avoiding excessive exposure to any single company and supporting compliance with SEBI's diversification and concentration requirements applicable to thematic equity schemes.
- The scheme therefore provides investors with diversified exposure to businesses that are expected to benefit from the long-term growth of education-related spending in India, while retaining a clear and measurable link to the underlying investment theme.

5.2 Total Expense Ratio

The scheme shall levy a Total Expense Ratio (TER) consistent with prevailing industry practice for thematic passive products. At launch, the expected TER is in the range of 0.45%-0.55% per annum.

TER component	Indicative (bps)
Index licensing fee (paid to the independent index provider, Section 5.4)	5-8
Fund management / AMC fee	25-30
Custody, RTA, trustee, audit	8-10
GST and other statutory costs	5-7
Indicative total TER	45-55

The TER may be revised from time to time within limits prescribed under applicable SEBI regulations.

5.3 Asset Scale and Operating Economics

The operating-cost structure of the ETF comprises index licensing, fund administration, custody, registrar and transfer agency services, trustee oversight, audit expenses and exchange-listing costs.
Based on prevailing market costs, annual fixed operating expenses are estimated at approximately ₹60-90 lakh. At a TER of 0.50%, the scheme achieves operating break-even at an estimated asset base of approximately ₹150 crore. As with most new thematic ETFs, scale is expected to build progressively through SIP flows, advisor distribution and institutional participation.

5.4 Creation and Redemption Mechanism

- The scheme shall employ the standard in-kind creation and redemption process applicable to ETFs in India.
- Authorized Participants (APs) shall create or redeem ETF units in Creation Unit sizes specified by the AMC. Each Creation Unit shall comprise the prescribed Portfolio Composition File (PCF), containing the constituent securities and weights determined under the Composite & Capped methodology described in Section 3.3. Portfolio weights shall be derived from a composite score based on Purity, Inflation Beta, and OPM Stability, subject to a maximum constituent weight of 15.00%, with excess weight redistributed across the remaining constituents.
- Retail investors shall buy and sell ETF units on recognized stock exchanges. Authorized Participants and market makers shall facilitate liquidity and arbitrage between the ETF's market price and its indicative Net Asset Value (iNAV).
- The portfolio shall be managed in accordance with a publicly disclosed methodology setting out constituent eligibility criteria, scoring and weighting rules, constituent caps, and semi-annual rebalancing and reconstitution procedures.

5.5 Portfolio Construction and Risk Controls

The portfolio incorporates a rules-based methodology designed to balance thematic purity, diversification, and investability.

- Constituent weights are determined using the Composite & Capped framework described in Section 3.3, which combines Purity, Inflation Beta, and OPM Stability into a single scoring mechanism. To mitigate concentration risk and maintain diversification, individual constituent weights are capped at 15.00%, with excess weight redistributed across the remaining portfolio.
- Constituents must demonstrate a documented linkage to one or more expenditure categories represented within the Education-CPI framework. Minimum liquidity and trading-frequency requirements are applied at each review to ensure investability and efficient portfolio implementation.
- The portfolio is rebalanced and reconstituted semi-annually to reflect changes in constituent fundamentals, inflation sensitivity, and eligibility criteria, while maintaining alignment with the education-inflation theme.

These controls are intended to preserve the portfolio's exposure to education-cost inflation while maintaining diversification, liquidity, and operational viability.

5.6 Investment Proposition

The Education Inflation ETF provides investors with a listed and transparent vehicle for gaining exposure to companies linked to the education expenditure ecosystem in India.
Unlike broad-market equity funds, the scheme follows a specialized portfolio methodology that selects and weights constituents based on their linkage to education-related spending, inflation sensitivity, and business stability. This offers investors a differentiated allocation aligned with the long-term growth of education costs.
The Education-CPI serves as the scheme's thematic benchmark, providing a reference measure for changes in education-related costs, while the ETF offers equity exposure to businesses positioned to benefit from those expenditure trends.

6. Conclusion

This paper demonstrates that education-cost inflation can be measured, indexed, and translated into an investable ETF using publicly available data, transparent methodology, and existing market infrastructure. The proposed Education-CPI Index combines education-related expenditure categories into a single benchmark, while the corresponding ETF provides exposure to listed companies whose revenues are linked to those costs.
The analysis shows that the concept is both operationally feasible and compatible with India's existing ETF and thematic-fund framework. No new regulation or proprietary dataset is required; the necessary information already exists across government statistics, industry sources and public-company disclosures.
The key insight is that the absence of an education-inflation-linked investment product in India is not a data problem but a product-design opportunity. The Vidyarth ETF provides a practical blueprint for aligning investment portfolios with one of the most significant long-term financial goals faced by Indian households.

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Annexure A: Explanation of Hedge-Effectiveness Metrics

A.1 Inflation Beta (3.2)

- Inflation Beta measures the sensitivity of portfolio returns to changes in the Education Consumer Price Index (Education-CPI).
- A value of 1.0 would indicate that portfolio returns move proportionately with education inflation. A value greater than 1.0 indicates that portfolio returns have historically increased at a faster rate than the inflation benchmark.
- The portfolio recorded an inflation beta of 3.2, implying that, during the observation period, each 1% increase in Education-CPI inflation was associated with an average increase of approximately 3.2% in portfolio returns. This figure was estimated using a regression analysis of annual portfolio returns against annual Education-CPI changes.

A.2 Coefficient of Determination ($R^2 = 0.28$)

R^2 measures the proportion of variation in portfolio returns that can be explained by movements in the Education-CPI benchmark.
The metric ranges from 0 to 1:

- 0 indicates no observable relationship.
- 1 indicates perfect explanatory power.

The portfolio's R^2 of 0.28 indicates that approximately 28% of annual return variation can be statistically explained by changes in education inflation. The remaining 72% is attributable to factors such as market conditions, company earnings, interest rates, investor sentiment, and other macroeconomic influences.

A.3 Maximum Drawdown (-15%)

Maximum Drawdown measures the largest decline in portfolio value from a historical peak to a subsequent trough before recovery.
The portfolio's maximum drawdown of -15% occurred during the COVID-19 market correction of 2020. This means that an investor holding the portfolio at its pre-crisis peak would have experienced a temporary decline of approximately 15% before the portfolio recovered.
Maximum drawdown is commonly used to assess downside risk and the magnitude of losses that may occur during periods of market stress.

A.4 Sharpe Ratio (0.49)

The Sharpe Ratio evaluates risk-adjusted returns by comparing excess returns to portfolio volatility.
A Sharpe Ratio of 0.49 indicates that the portfolio generated approximately 0.49 units of excess return for every unit of risk undertaken during the analysis period.

A.5 Relative Shortfall (23.7%)

- Relative Shortfall measures the maximum gap between the portfolio's cumulative value and the cumulative Education-CPI benchmark during the study period.
- The largest shortfall occurred during the 2020 market correction, when the portfolio declined while education inflation continued to rise. At its widest point, the portfolio lagged the inflation benchmark by approximately 23.7 percentage points.
- This metric provides an indication of the extent to which the inflation hedge may temporarily weaken during severe market disruptions, even if long-term performance remains favourable.